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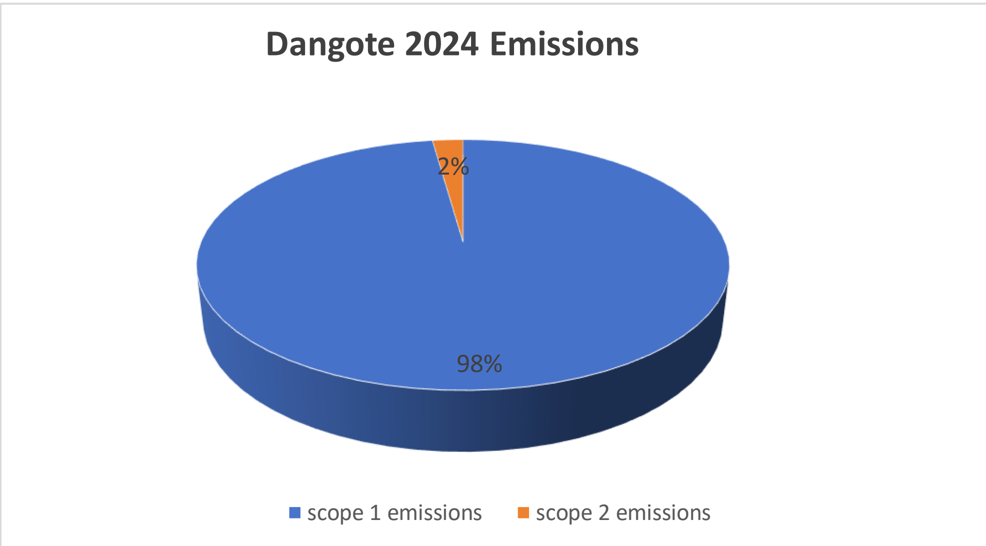
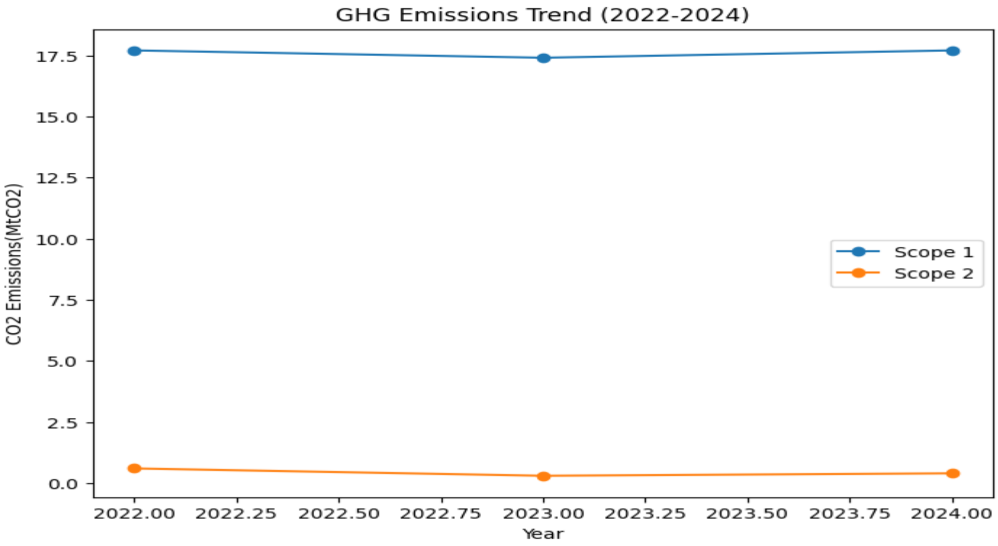
DANGOTE SUSTAINIBILITY INSIGHTS AND DECARBONIZATION ROADMAP

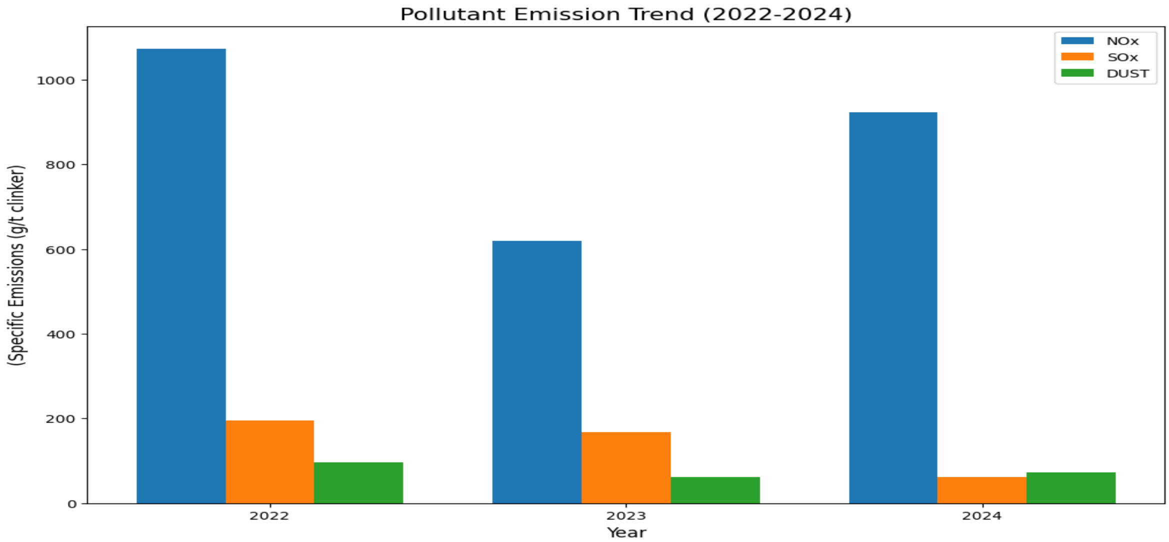
The Dangote Cement sustainability insights and decarbonization roadmap examines the transition of cement towards lower carbon intensity through improved efficiency, alternative materials and cleaner production technologies. As one of Africa's largest cement producers, dangote cement plays a critical role in addressing industrial emissions, particularly through clinker substitution and energy optimization strategies. This analysis highlights the need for sustainable innovation in the cement sector and explores practical pathways for reducing environmental impact while maintaining production growth and economic competitiveness.

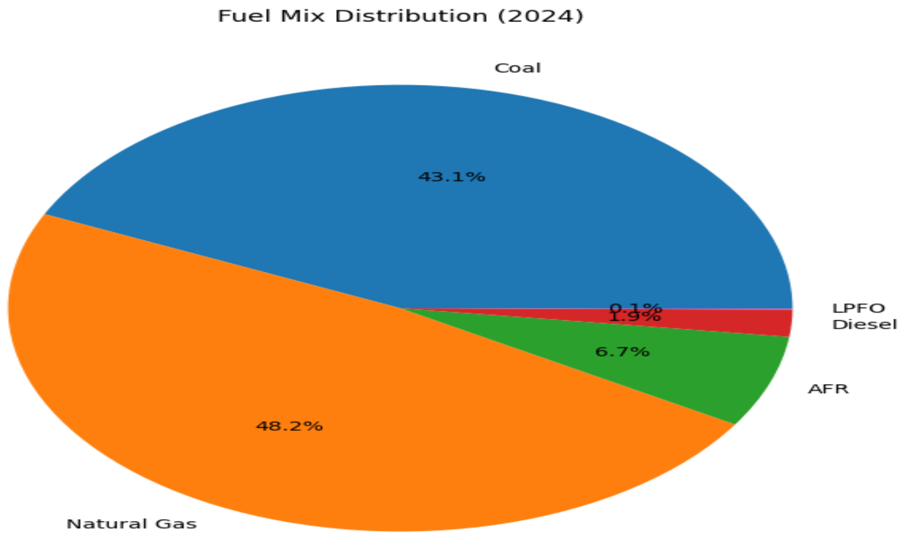
The Data set was extracted from Dangote cement sustainability report 2024. Graphs and charts were plotted with the aid of Python Matplotlib and WPS sheets.

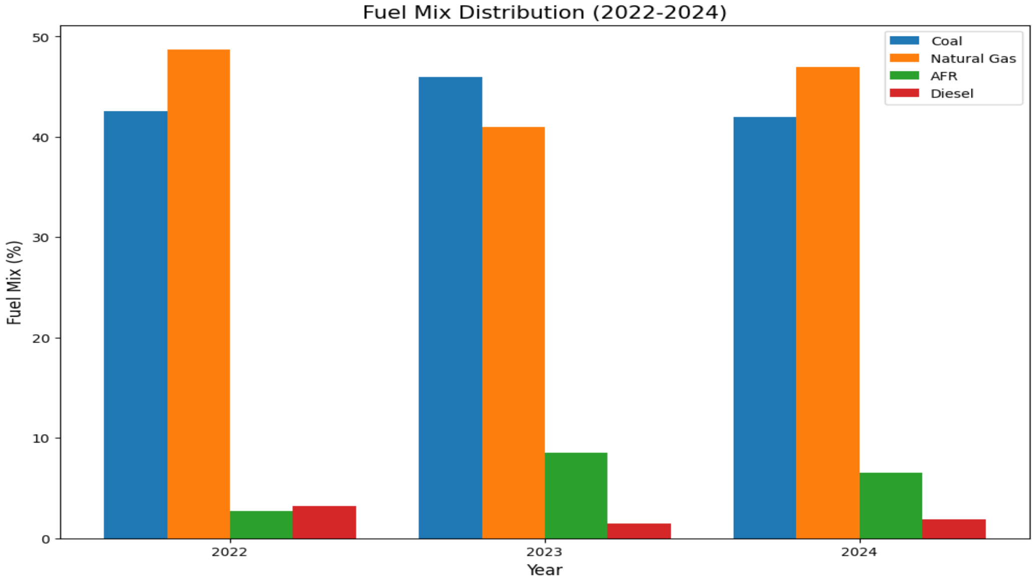
DATA VISUALIZATION




GHG Emissions Trend (2022-2024)
CO2 Emissions(MtCO2)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2022.00
2022.25
2022.50
2022.75
2023.00
2023.25
2023.50
2023.75
2024.00
Year
Scope 1
Scope 2


Pollutant Emission Trend (2022-2024)
(Specific Emissions (g/t clinker)
1000
800
600
400
200
0
2022
2023
2024
Year
NOx
SOx
DUST


Fuel Mix Distribution (2024)
Coal
43.1%
LPFO
Diesel
0.1%
1.9%
6.7%
AFR
48.2%
Natural Gas



Indicator	2022	2023	2024	Trend	
Scope 1 emissions	17.7	17.4	17.7	Relatively stable	Scope 1 emissions remained relatively stable from the base year suggesting that on-site power generation, fossil fuel(coal, natural gas) and clinker production dominates dangote cement GHG emissions
Scope 2 emissions	0.3	0.6	0.4	Fluctuating	Scope 2 emissions fluctuated from base year with a noticeable increase in 2023 followed by a reduction in 2024. this variation is directly linked to the percentage of plant electricity consumption. The initial increase might be as a result of the creation of new offices which requires more load or even the purchase of new equipment. The decrease seen in 2024 might be as a result of

					purchasing more energy efficient equipment or reducing usage. another angle is that these changes can be attributed to grid emission factors from electricity suppliers which dangote cement has no control over.
Net CO2 per tonne of cementititious material(kgCO2/tonne)	590	577	570	Decreasing	The reduction in net C02 intensity suggests improvements in decarbonization efforts across the production system. But more work has to be done as Environmental pressure mounts.
Specific heat consumption (MJ/tonne clinker)	3,330	3,428	3,281	Improved in 2024	The decrease in specific heat consumption in 2024 indicates improved kiln efficiency which contributes to lower fuel demand/usage and reduced process emissions intensity as can be seen in the Net CO2 per tonne of cementititious material(kgCO2/tonne) above
Alternative fuels and Raw materials(AFR) (%)	2.7	8.5	6.5	Increased significantly	The increased utilization of Alternative fuels and raw materials(AFR) demonstrates growing adoption of circular economy.
SOx(g/tonne)	195	168	61	Sharp Decrease	The substantial reduction in SOx from the base year may suggest improved fuel quality as lower fuel quality has the tendency to emit more SOx. Some sulfur control measures might also have been put in place alongside the optimization of the combustion process. This was measured using

					CEMS (Continuous Emissions Monitoring Systems) used in the facility. This shows the quality of the data provided as it is the highest quality data in Greenhouse gas accounting.

DECARBONIZATION PATHWAY

Energy usage

The Emission reduction scenario was developed based on the assumption that the increase in the utilization of alternative fuels and raw materials(6.5% to 20% of plant energy use) would partially substitute fossil fuel consumption within the clinker production process. A hypothetical 10% reduction in scope 1 emissions was therefore considered to evaluate the potential contribution of AFR adoption towards decarbonization.

A 10% reduction in scope 1 emissions : 17.7(1 - 0.10) = 15.93MtCO2

Among the potential biomass derived AFR options in Nigeria, palm kernel shells, rice husk and bagasse demonstrate significant potential due to their abundance, renewable origin and calorific value. But palm kernel and rice husk supplies are usually fragmented as they spread widely across the country. Since bagasse is a waste from dangote sugar refinery, it can be sun dried to reduce moisture content, pelletized to Increase bulk density for cost effective transportation logistics. For this to occur, its kiln compatibility has to be tested for a possible industrial scale integration.

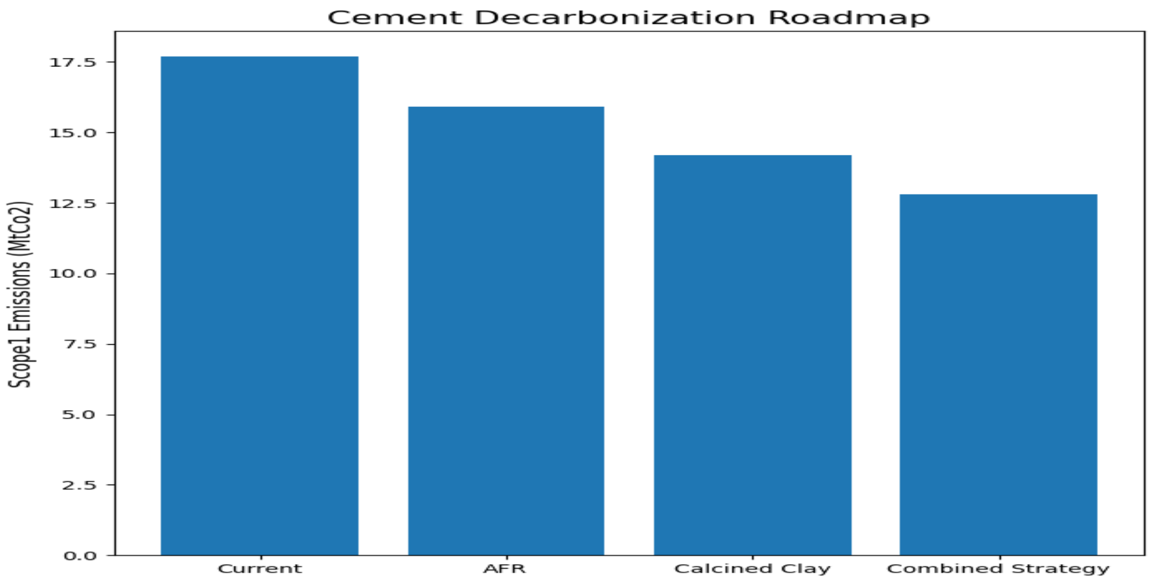
Countries with established sugar industries such as Brazil and India have successfully adopted bagasse into co-firing systems to reduce fossil fuel usage and greenhouse gases.

Clinker substitution

Clinker substitution through calcined clay and LC3 cement represents one of the most promising decarbonization pathways for the cement industry due to its ability to directly to reduce process emissions associated with clinker production. Dangote cement's participation in the Tanzania cement Industry Net Zero Accelerator Roadmap and Congo's approval of lower clinker factor standards for LC3 cement highlights the growing industrial relevance of calcined clay integration within Africa. Considering Nigeria's abundant kaolin deposits in Ogun, Edo, Kogi, Plateau, Kaduna and Katsina. Local cement manufacturers should accelerate the adoption of calcined clay-based cement systems to reduce clinker dependence, lower carbon emissions and support sustainable industrial development. This strategy is in line with Nigeria's NDC 3.0 under the paris agreement by substituting 10% clinker in 33% of cement.

One of the major challenges associated with clinker substitution adoption is the significant capital already invested by cement manufacturers such as dangote cement in conventional clinker infrastructure . Large-scale reduction in clinker production may therefore present economic and operational concerns. However, strategic partnerships with external companies specialized in kaolin calcination and metakaolin production provides a more feasible pathway for LC3 integration without requiring complete modification of existing kiln systems.

A 20% reduction in clinker by substituting with calcined clay : 17.7(1 - 0.20) = 14.16MtCO2



Conclusion

Cement decarbonization involves a combined strategy pathway involving alternative fuels, clinker substitution and process optimization. Furthermore, the adoption of solar energy may enable the company earn international Renewable Energy Certificates (I-RECs), which can be accounted for under the market-based approach of the GHG Protocol Scope 2 reporting framework. While scope 2 emissions contribute less significantly to the overall cement carbon footprint compared to scope 1 process emission, renewable electricity integration still supports broader corporate sustainability goals, ESG performance improvement and transition toward low-carbon industrial operations.